FIITJEE Students From Classroom / Integrated School Programs have secured to 2 Zonal, 6 State & 18 City Topper Ranks. 33 in Top 100, 78 in Top 200 and 205 in Top 500 All India Ranks bagged by FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

ALL INDIA TEST SERIES

FIITJEE

JEE (Advanced)-2016

FULL TEST – II

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 243**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section – A (01 – 06)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

   Section – A (07 – 09) contains 3 multiple choice questions which have one or more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

   Section-A (10 – 14) contains 2 paragraphs. Based upon paragraph, 2 and 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.
2. **Section – B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 marks** will be awarded. There may be one or more than one correct matching. No marks will be given for any wrong matching in any question. There is no negative marking.
3. **Section-C (01 – 05)** contains 5 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

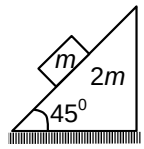
SECTION – A

Straight Objective Type

This section contains 6 multiple choice questions numbered 1 to 6. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. Which of the following is the most accurate instrument for measuring length?
   (A) vernier calipers having 20 divisions on the sliding scale which coincide with 19 divisions on the main millimeter scale
   (B) a screw gauge having pitch 1 mm and 50 divisions on the circular scale
   (C) a vernier scale of least count 0.01 mm
   (D) a screw gauge of least count 0.001 mm

2. A block of mass *m* is placed on a wedge of mass 2m which rests on a rough horizontal surface. There is no friction between the block and the wedge. The minimum coefficient of friction between the wedge and the ground so that the wedge does not move is



   (A) 0.167 (B) 0.20
   (C) 0.23 (D) 0.33

3. A small ball thrown with an initial velocity *u* directed at an angle $\theta = 37°$ above the horizontal collides inelastically ($e = 1/4$) with a vertical massive wall moving with a uniform horizontal velocity $u/5$ towards ball. After collision with the wall, the ball returns to the point from where it was thrown. Neglect friction between ball and wall. The time *t* from beginning of motion of the ball till the moment of its impact with the wall is ($\tan 37° = 3/4$)
   (A) $\frac{3u}{5g}$ (B) $\frac{18u}{25g}$
   (C) $\frac{54u}{125g}$ (D) $\frac{54u}{25g}$

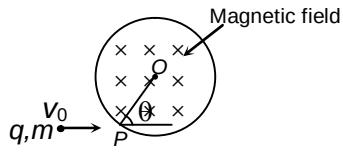
Space for Rough work

4. A cube of side *l* and mass *M* is placed on rough horizontal surface and the friction is sufficient so that it will not move, if a constant force *F* = *Mg* is applied horizontally *l*/4 above the surface. Then the torque due to normal force about center of the cube is equal to

(A) $\frac{Mgl}{2}$ (B) $\frac{Mgl}{4}$

(C) $\frac{Mgl}{8}$ (D) zero

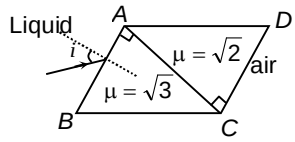
5. A particle of charge *q* and mass m is projected with a velocity v_0 towards a circular region having uniform magnetic field *B* perpendicular and into the plane of paper from point *P* as shown in the figure. *R* is the radius and *O* is the centre of the circular region. If the line OP makes an angle θ with the direction of v_0 then the value of v_0 so that particle passes through *O* is



(A) $\frac{qBR}{m\sin\theta}$ (B) $\frac{qBR}{2m\sin\theta}$

(C) $\frac{2qBR}{m\sin\theta}$ (D) $\frac{3qBR}{2m\sin\theta}$

6. A ray of light from a liquid $\left(\mu=\sqrt{3}\right)$ is incident on a system of two right angled prism of refractive indices $\sqrt{3}$ and $\sqrt{2}$ as shown. The ray suffers zero deviation when emerges into air from *CD*. The angle of incidence *i* is



(A) 45^0 (B) 35^0

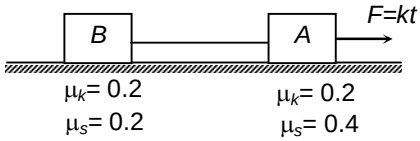
(C) 20^0 (D) 10^0

Space for Rough work

Multiple Correct Choice Type

This section contains 3 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

7. Two blocks *A* and *B* each of mass $\frac{1}{2}$ kg is connected by a massless inextensible string and kept on horizontal surface. Coefficient of friction between block and surface is shown in figure. A force $F = kt$ (where *k* = 1 N/s and *t* is time in second) applied on *A*. Then (g = 10 m/s^2)



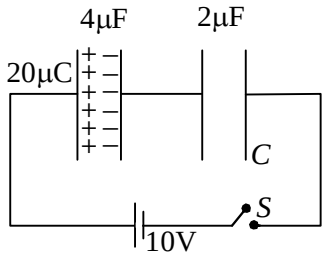
(A) work done by friction force on block *B* is zero in time interval *t* = 0 to *t* = 3s.
(B) work done by friction force on block *A* is zero in time interval *t* = 0 to *t* = 3s.
(C) work done by tension on *B* is also zero in time interval *t* = 0 to *t* = 3s.
(D) speed of blocks at *t* = 10s is 27.5 m/s.

8. A particle starts SHM at time $t = 0$. Its amplitude is *A* and angular frequency is ω. At time $t = 0$ its kinetic energy is $\frac{E}{4}$, where *E* is total energy. Assuming potential energy to be zero at mean position, the displacement-time equation of the particle can be written as

(A) $x = A\cos\left(\omega t + \frac{\pi}{6}\right)$ (B) $x = A\sin\left(\omega t + \frac{\pi}{3}\right)$

(C) $x = A\sin\left(\omega t - \frac{2\pi}{3}\right)$ (D) $x = A\cos\left(\omega t - \frac{\pi}{6}\right)$

9. A $4\,\mu F$ capacitor is given $20\,\mu C$ charge and is connected with an uncharged capacitor of capacitance $2\,\mu F$ as shown in figure. When switch *S* is closed.



(A) charged flown through the battery is $\frac{40}{3}\mu C$

(B) charge flown through the battery is $\frac{20}{3}\mu C$

(C) work done by the battery is $\frac{200}{3}\mu J$

(D) work done by the battery is $\frac{100}{3}\mu J$

Space for Rough work

Comprehension Type

This section contains 2 paragraphs. Based upon one of paragraph 2 multiple choice questions and based on the other paragraphs 3 multiple choice questions have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Paragraph for Question Nos. 10 and 11

A massive disc of radius *R* is mounted on a light axle with the axle held horizontal, the disc is made to spin with angular speed ω_0. It is then lowered gently on to a level table and released as soon as its rim makes contact with the table top. The disc begins to move along the table, skidding and picking up speed as it goes.

10. If μ_k is the coefficient of kinetic friction between the rim of the disc and table top, disc will continue to skid for a time

(A) $\frac{R\omega_0}{3\mu_k g}$ (B) $\frac{R\omega_0}{2\mu_k g}$

(C) $\frac{R\omega_0}{\mu_k g}$ (D) $\frac{3R\omega_0}{\mu_k g}$

11. If initial kinetic energy of the disc is *K*, then the kinetic energy of the disc after it stars pure rolling will be

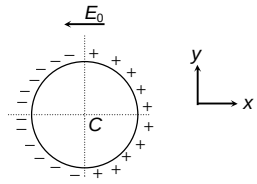
(A) $\frac{2}{3}K$ (B) $\frac{1}{2}K$

(C) $\frac{1}{3}K$ (D) $\frac{3}{4}K$

Space for Rough work

Paragraph for Question Nos. 12 and 14

A uniform ring of mass m and radius R can rotate freely about an axis passing through centre C and perpendicular to plane of paper. Half of ring is positively charge and other half is negatively charge. Uniform electric field E_0 is switched on along –ve x-axis (Axis are shown in figure)
[magnitude of charge density λ]



12. The dipole moment of ring is
(A) $2\lambda R^2$ (B) $4\lambda R^2$
(C) $2\pi\lambda R^2$ (D) $4\pi\lambda R^2$

13. The equilibrium of ring is
(A) stable (B) unstable
(C) neutral (D) can't say

14. If ring is slightly disturb from given position, find the angular speed of ring when it rotate by $\pi/2$.
(A) $2\sqrt{\frac{\lambda E_0}{m}}$ (B) $\sqrt{\frac{\lambda E_0}{m}}$
(C) $\sqrt{\frac{8\lambda E_0}{m}}$ (D) none

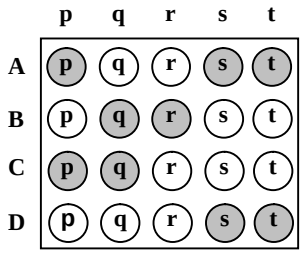
Space for Rough work

SECTION – B

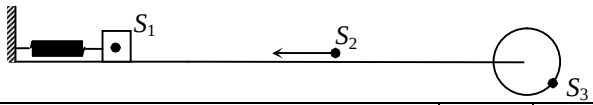
(Matrix-Match Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. Three sound sources of same frequency f are arranged as shown in figure. S_2 moves towards left with speed $\frac{v}{n}$, S_3 moves on a circular path of radius R with constant speed $\frac{2v}{n}$, and S_1 performs S.H.M. with frequency $\frac{100v}{n\pi}s^{-1}$ and amplitude 1 cm. ($n > 12$ and v is speed of sound)



	Column I		Column II
(A)	Beat frequency heard by S_1 may be	(p)	$\frac{f}{n}$
(B)	Beat frequency heard by S_2 may be	(q)	$\frac{2f}{n+1}$
(C)	Beat frequency heard by S_3 may be	(r)	zero
(D)	Change in the wave length of wave emitted by S_3 and observed by S_2 may be	(s)	$\frac{2f}{n-1}$
		(t)	

Space for Rough work

2. Match the following:

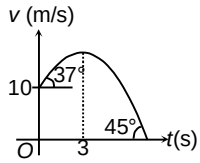
	Column I		Column II
(A)	The voltage applied to *X*-ray tube is increased.	(p)	Average K.E. of the electrons decreases
(B)	In photo electric effect work function of the target is increased.	(q)	Average K.E. of the electrons increases
(C)	Stopping potential decreased by	(r)	Cut off wavelength decreases
(D)	Wavelength of K_α , *X*-ray increased by	(s)	Atomic number of target material decreases
		(t)	

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. A particle starts moving with velocity 10 m/s in a straight line under an acceleration varying linearly with time. Its velocity time graph is as shown in figure. Its velocity is maximum at t = 3 sec. Find the time (in sec) when the particle stops (tan 37^0 = ¾)



2. A block of mass 1 kg start moving at t = 0 with speed 2 m/s on rough horizontal surface with coefficient of friction 0.2. A horizontal force F is applied in the direction of velocity which varies with time shown in figure (B). Find the speed (in m/s) of particle at t = 3 s (g = 10 m/s^2).

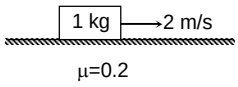
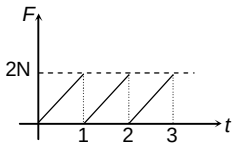


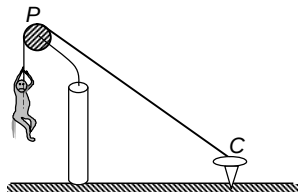
Figure (a)



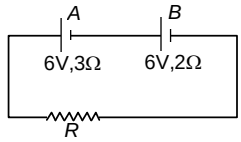
Figure

Space for Rough work

3. One end of a massless rope, which passes over a massless and frictionless pulley *P* is tied to a hook *C* while the other end is free. Maximum tension that the rope can bear is 360 N with what value of minimum safe acceleration (in ms^{-2}) can a monkey of 60 kg move down on the rope



4. Two sources of emf 6V and internal resistance 3Ω and 2Ω are connected to an external resistance *R* as shown. If potential difference across battery *A* is zero, then find the value of *R* (in ohm)



5. Two radioactive elements *R* and *S* disintegrate as

$R \longrightarrow P + \alpha$; $\lambda_R = 4.5 \times 10^{-3}$ years $^{-1}$

$S \longrightarrow Q + \beta$; $\lambda_S = 3 \times 10^{-3}$ years $^{-1}$

Starting with number of atoms of *R* and *S* in the ratio of 2 : 1, find the ratio after the lapse of three half lives of *R*……..

Space for Rough work

Chemistry

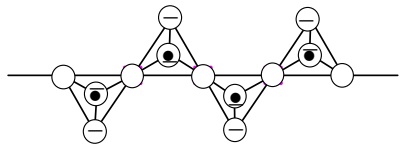
PART – II

SECTION – A

Straight Objective Type

This section contains 6 multiple choice questions numbered 1 to 6. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. The formula of the following chain silicate (pyroxene) can be represented as



(A) $(Si_3O_9)^{6-}$ (B) $(SiO_4)_n^{3n-}$

(C) $(SiO_3)_n^{3n-}$ (D) $(SiO_3)_n^{2n-}$

2. Which of the following equations must be used for the exact calculation of $[H^+]$ of an aqueous HCl solution at any concentration C_{HCl}. $[K_w = 10^{-14} M^2]$

(A) $[H^+] = C_{HCl}$ (B) $\left[H^+\right] = C_{HCl} + K_w$

(C) $\left[H^+\right] = C_{HCl} + \frac{K_W}{\left[H^+\right]}$ (D) $\left[H^+\right] = C_{HCl} - \frac{K_w}{\left[H^+\right]}$

3. Which of the following salts give different result by the action of heat?

(A) Li_2CO_3 and $CaCO_3$ (B) $NaHCO_3$ and $KHCO_3$

(C) Hydrated $MgCl_2$ and hydrated Na_2SO_4 (D) Li_2SO_4 and $MgSO_4$

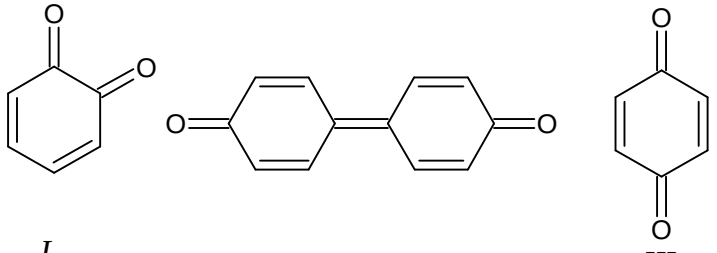
4. $A \xrightarrow{High\ P\&T} 3B + C$ is a first order reaction. If A is strong mono protic acid and B is strong diprotic acid and reaction kinetic is studied by using a standard solution of NaOH required to neutralise reaction mixture at different instants of time. When initially we had taken same concentration of A in each case.
Now it was observer that volume required after 971 sec was double that required initially, then the half life of reaction is:

(A) 485 sec (B) 1942 sec

(C) 3010 sec (D) 971 sec

Space for Rough work

5. Consider the following conjugated diketo compounds:



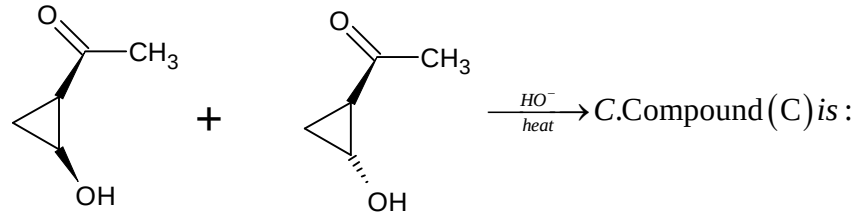
I *II* *III*

Which of reduction potential the following is the correct decreasing order of $E°$ values:

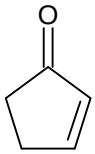
(A) $I > II > III$ (B) $III > II > I$

(C) $I > III > II$ (D) $II > I > III$

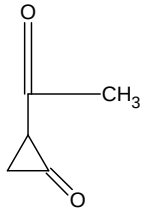
6. Compound A and B, both were treated with $NaOH$, producing a single compound C:



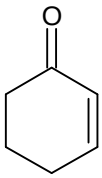
(A)



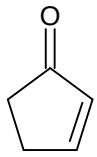
(B)



(C)



(D)



Space for Rough work

Multiple Correct Choice Type

This section contains 3 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

7. For the reaction $NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g)$. If Kp of the above reaction is 78.1 at $400°C$ which is not correct:

(A) At pressure P degree of dissociation of NH_3 is $\left[1+\frac{3\sqrt{3}P}{4k_p}\right]^{-1/2}$

(B) $K_c = 1.413\, mol/\ell$

(C) If we increase pressure it will decrease K

(D) If we increase P by bringing piston down , then, degree of dissociation α will decrease

8. The correct statement(s) about solvent effect is/are

(A) Decreasing solvent polarity causes a large increase in the rate of the S_{N^2} attack by ammonia on an alkyl halide $R-X+\ddot{N}H_3 \longrightarrow R\overset{+}{N}H_3 + X^-$

(B) Increasing solvent polarity causes a large decrease in the rate of the S_{N^2} attack by hydroxide ion on trimethyl sulphonium ion.

$$HO^- + \underset{Trimethyl\ \ sulphonium\ ion}{(CH_3)_3S^+} \longrightarrow CH_3OH + \underset{DMS}{(CH_3)_2S}$$

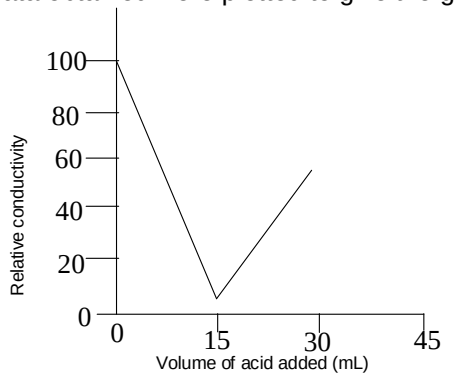
(C) Increasing solvent polarity causes a small decrease in the rate of the S_{N^2} attack by trimethyl amine on trimethyl sulphonium ion

$$(CH_3)_3\ddot{N} + (CH_3)_3\overset{+}{S} \longrightarrow CH_3\overset{+}{N}(CH_3)_3 + (CH_3)_2S$$

(D) All are incorrect.

Space for Rough work

9. 20 mL of KOH solution was titrated with 0.20 mol L^{-1} H_2SO_4 solution in a conductively cell. The data obtained were plotted to give the graph shown below:



The concentration of the KOH solution was:

(A) 0.30 mol L^{-1} (B) 0.15 mol L^{-1}

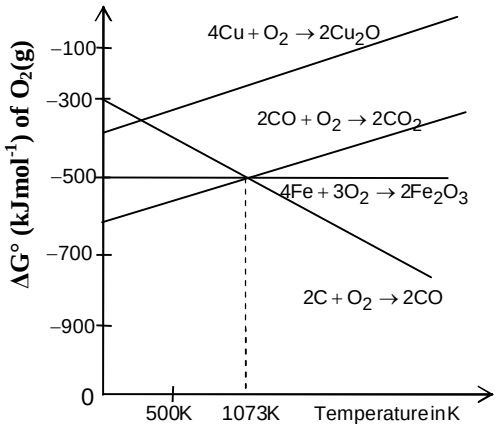
(C) 0.12 mol L^{-1} (D) 0.075 mol L^{-1}

Comprehension Type

This section contains 2 paragraphs. Based upon one of paragraph 2 multiple choice questions and based on the other paragraphs 3 multiple choice questions have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Paragraph for Question Nos. 10 and 11

The Ellingham diagram for the reduction of haematite and that of cuprous oxide given below;



On the basis of the above plot, answer the following two questions:

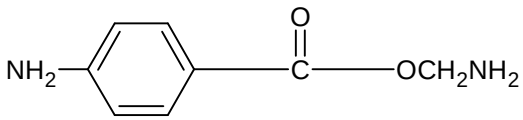
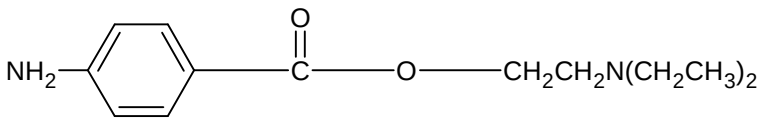
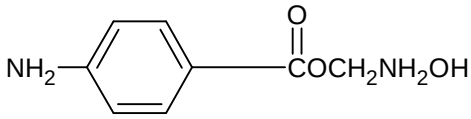
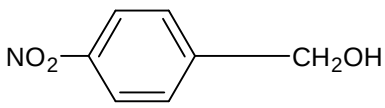
Space for Rough work

10. Choose the incorrect statement
(A) Above 1073 K carbon can reduce Fe_2O_3
(B) Below 1073 K ΔG° of the reaction given below is negative
$Fe_2O_3(s) + 3CO(s) \longrightarrow 2Fe(s) + 3CO_2(g)$
(C) Below 1073 K ΔG° of formation of CO_2 from CO is more negative than ΔG° of formation of Fe_2O_3.
(D) Above 1073 K ΔG° for the formation of CO is less negative than that of Fe_2O_3.

11. The incorrect statement amongst the following is
(A) The ΔG° of the formation of CO_2(g) will remains almost independent of temperature because of ΔS° of the reaction being almost equal to zero.
(B) The reaction: 2C(s) + O_2(g) → 2CO(g) is most suitable at temperature above 1073 K.
(C) It is difficult to reduce Cu_2O to the metal by heating with coke as compared to Fe_2O_3.
(D) Reduction with carbon at high temperature is not preferred due to carbide formation.

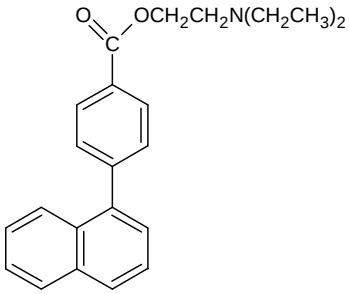
Paragraph for Question Nos. 12 and 14

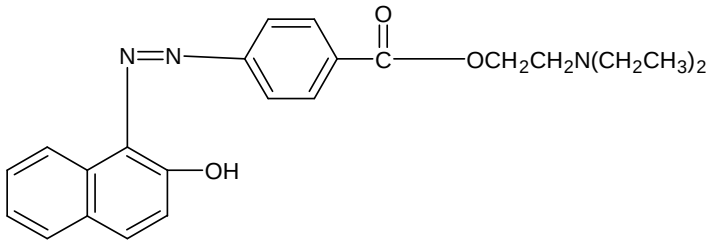
Dolacaine, a local anesthetic, is a compound with molecular formula, $C_{13}H_{20}O_2N_2$. It is insoluble in water and dilute NaOH, but soluble in dil. HCl. Upon treatment with $NaNO_2$ and HCl and then with β -naphthol, a highly coloured solid is formed. When Novocaine is boiled with aqueous $NaOH$, it slowly dissolves. The alkaline solution is shaken with ether and layers are separated.
Acidification of the aqueous layer causes the precipitation of white solid 'A'; continued addition of acid causes 'A' to redissolve. Upon isolation 'A' is found to have melting point of 185 -186° C and the formula $C_7H_7O_2N$:
Evaporation of ether layer leaves a liquid 'B' of formula $C_6H_{15}ON$. with acetic anhydride gives 'C', $C_8H_{17}O_2N$ which is insoluble in water and dilute bases, but soluble in dilute HCl.
'B' is found to be identical to the compound formed by the action of diethylamine on ethylene oxide.

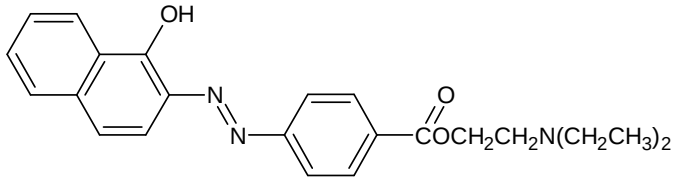
12. When compound 'B' is treated with p – nitrobenzoyl chloride followed by moderate reduction with Ni/H_2; the compound formed is:
(A) 
(B) 
(C) 
(D) 

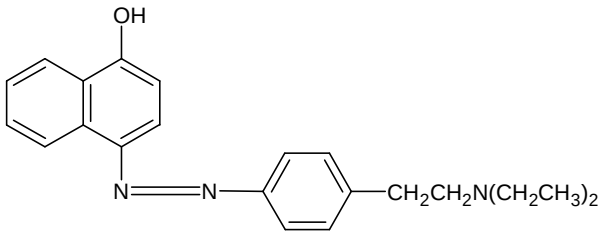
Space for Rough work

13. Dolacaine $\xrightarrow{NaNO_2, HCl}$ $\xrightarrow{\beta-naphthol}$ Product

(A) 

(B) 

(C) 

(D) 

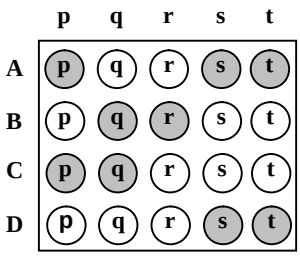
Space for Rough work

14. Compound B is having:
(A) 3° amine (B) ether group
(C) acid group (D) nitro group

SECTION – B

(Matrix-Match Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:
If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. [Use $\log 1.8 = 0.26$, ka of formic acid $= 1.8\times10^{-4}$ for acetic acid $= 1.8\times10^{-5}$, K_a of $NH_3 = 1.8\times10^{-5}$ Ka_1, of $H_2S = 10^{-7}$ and Ka_2 of $H_2S = 10^{-14}$]

Match the entries of column – II for which the equality or inequality given in the column – I is valid:

Column – I		Column – II	
(A)	$10^{-5}\,M\ HCl$ solution> $0.1M\ H_2S$ solution	(p)	α_{water} (degree of dissociation of water)
(B)	CH_3COOH solution at pH equal to $4.74 = NH_4OH$ solution at pH equal to 9.26	(q)	$\left[O\bar{H}\right]$
(C)	$0.1M$ CH_3COOH solution $= 1.0M$ $HCOOH$ solution	(r)	$\alpha_{electrolyte}$ (degree of dissociation of weak electrolytes)
(D)	$0.1M$ of a weak acid $HA\left[K_c = 10^{-5}\right]$ solution $< 0.01M$ of a weak acid $HA_2\left[K_a = 10^{-8}\right]$ solution	(s)	pH
		(t)	Essentially acidic solution

Space for Rough work

2. Matrix type the following questions:

Column – I (Cell)		Column – II (EMF or Electrode potential)	
(A)	Hydrogen electrode, reduction potential, set up in 1.0 M KCl, pH_2 = 1 atm	(p)	$1.24V$
(B)	$Pt\left(H_2\ at\ 1atm\right)\ \left\|H^+\left(pH=2\right)\right.$	(q)	$-0.414V$
(C)	$Hg\ \left\|Hg_2Cl_2\ \right\|Cl^-\left(0.1M\right)\ \left\|\right\|Cl^-\left(0.01M\right)\ \left\|Pt\left(Cl_2\right)\right.$ $\left(E^\circ=1.1\ V\right)$	(r)	$0.118\ V$
(D)	Reduction potential: $AgI\left(aq\right)$ saturated $\left\|\ Ag\right.$ $\left(E°_{Ag^+/Ag}=0.8\ V, Ksp\ \text{of}\ AgI=10^{-16}\right)$	(s)	$0.327V$
		(t)	Metal-metal insoluble salt electrode

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. Sulphide ion react with $Na_2[Fe(NO)(CN)_5]$ to form a purple coloured compound $Na_4[Fe(CN)_5NOS]$. In this reaction the change in oxidation state of iron is?

2. Gold number of starch is 25. The amount of $0.3125\times a$ gm of starch is required to prevent coagulation of 100 ml of gold sol adding 10 ml of 10% NaCl solution? Value of a is…….

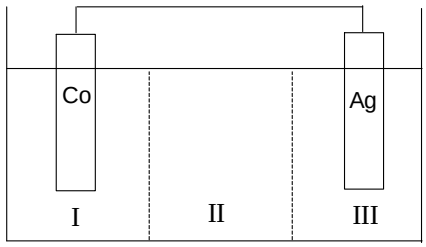
3. Dimethyl gloyoxime is added to aqueous solution of nickel chloride, made alkaline, by adding NH_4OH solution, a brilliant red ppt is obtained. The number of $Ni-N$ bonds in red ppt is……..?

Space for Rough work

4. A coordination compound has empirical formula PdF_3. What is oxidation number of central atom?

Comprehension – (Question No. 5)

The cell given below consists of three compartments separated by porous barriers. The first contains a cobalt electrode in 5 L of 0.1 M $Co(NO_3)_2$; the second contains 5 L of 0.1 M KNO_3; the third contains a silver electrode in 5 L of 0.1 M $AgNO_3$. Assume that the current within in the cell is carried equally by the positive and negative ions(does not depend on the size) if there is a passage of 0.1 mole of electrons.



$E^{\circ}_{Co^{2+}/Co} = -0.28\ V$

$E^{\circ}_{Ag^{+}/Ag} = 0.8\ V$

5. The concentrations of Co^{2+} in the second compartment is $\frac{x}{1000}$ M, then the x will be:

Space for Rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains 6 multiple choice questions numbered 1 to 6. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. $\lim_{x\to 0}\frac{\sin^{-1}x-\tan^{-1}x}{x^3}$ is equal to

(A) $\frac{1}{6}$ (B) $\frac{1}{2}$

(C) 2 (D) $-\frac{1}{2}$

2. Area of the region bounded by the curve $y=\frac{1}{x}+1,$ the tangent at $P(2,\frac{3}{2})$ and the line $x=1$ is

(A) $\frac{4}{5}+\ell n2$ (B) $-\frac{5}{4}+\ell n3$

(C) $-\frac{5}{8}+\ell n2$ (D) $\ell n2-\frac{5}{8}$

3. Solution of the differential equation $\frac{dy}{dx}=\frac{3x^2y^4+2xy}{x^2-2x^3y^3}$ is

(A) $x^2y^3+\frac{x}{y^2}=c$ (B) $x^3y^2+\frac{x^2}{y}=c$

(C) $\frac{x^2}{y^3}+xy^2=c$ (D) $x^3y^2-\frac{x^2}{y}=c$

Space for Rough work

4. Angle between $\bar{a}$ & $\bar{b}$ is $\frac{\pi}{4}$; $|\bar{a}| = 2\sqrt{2}$, $|\bar{b}| = 3$. Length of longer diagonal of the parallelogram having adjacent sides represented by $5\bar{a} + 2\bar{b}$ and $\bar{a} - 3\bar{b}$ is

(A) $\sqrt{593}$ (B) $\sqrt{26}$
(C) 15 (D) none of these

5. A number is selected at random from the set of natural numbers. The probability that the sum of the digits of its square is 39 is

(A) $\frac{1}{6}$ (B) 0
(C) $\frac{1}{10}$ (D) $\frac{1}{9}$

6. $A + B = \frac{\pi}{3}$; $\cos A + \cos B = 1$; value of $|\cos A - \cos B|$ is

(A) $\frac{1}{3}$ (B) $\sqrt{\frac{2}{3}}$
(C) $\sqrt{\frac{3}{2}}$ (D) $\frac{1}{2}$

Multiple Correct Choice Type

This section contains 3 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

7. If k denotes the remainder on dividing $\sum_{r=1}^{200} r!$ by n then (k, n) can be

(A) (5, 14) (B) (1, 3)
(C) (3, 10) (D) (3, 15)

Space for Rough work

8. The square roots of the matrix $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ are

(A) $\pm\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ (B) $\pm\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

(C) $\pm\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (D) $\pm\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

9. Range of the function $f(x) = \sqrt{\frac{1-|x|}{2-|x|}}$ contains

(A) $[0,1)$ (B) $[1,2)$

(C) $(1,2)$ (D) $(2,\infty)$

Comprehension Type

This section contains 2 paragraphs. Based upon one of paragraph 2 multiple choice questions and based on the other paragraphs 3 multiple choice questions have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Paragraph for Question Nos. 10 and 11

a, b, c, d is an A.P. with integer terms and $d = a^2 + b^2 + c^2$.

10. The common difference is

(A) 1 (B) 2

(C) 3 (D) none

11. The value of $a + 2b + 3c + 4d$ is

(A) 0 (B) 10

(C) 100 (D) none of these

Paragraph for Question Nos. 12 and 14

$f(x) = \sin 2\pi x + \{x\} : x \in [0,10]$.

12. Number of points where *f* achieves local maximum is

(A) 10 (B) 11

(C) 20 (D) none of these

Space for Rough work

13. Number of roots of $f(x) = 0$ in (0, 10) is

(A) 20 (B) 30
(C) 31 (D) none of these

14. Number of points where f achieves local minima is

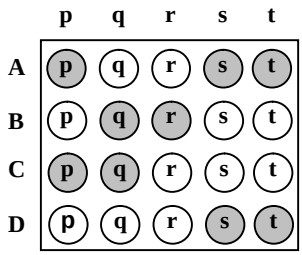
(A) 10 (B) 11
(C) 15 (D) none of these

SECTION – B

(Matrix-Match Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. Match the following column:

	Column I		Column II
(A)	Length of side of equilateral triangle with vertex at origin and inscribed in the parabola $y^2 = 12x$ is	(p)	$18\sqrt{3}$
(B)	Length of chord through origin of parabola $y^2 = 12x$ and having slope 1 unit is	(q)	$24\sqrt{3}$
(C)	Length of normal chord of the parabola $y^2 = 12x$ subtending a right angle at focus is	(r)	$7\sqrt{5}$
(D)	Length of normal chord of the parabola $y^2 = 12x$ subtending a right angle at the vertex is	(s)	$12\sqrt{2}$
		(t)	$15\sqrt{5}$

Space for Rough work

2. Consider the family of lines $ax+by+c=0$ where $4a^2+9b^2-c^2+12ab=0$. Let 'p' denote distance of R(6, 8) from the line $ax+by+c=0$

Match the following columns:

	Column I (value of p)		Column II (No. of lines)
(A)	$p=\sqrt{41}$	(p)	1
(B)	$p=2/\sqrt{13}$	(q)	2
(C)	$p=\sqrt{185}$	(r)	3
(D)	$p=\sqrt{80}$	(s)	4
		(t)	5

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. Number of points of discontinuity of the function $f(x)=\left[x^2+1\right]; x\in[1,3]$ is

2. A tangent to ellipse $x^2+4y^2=4$ meets the ellipse $x^2+2y^2=6$ at P and Q. Tangents at P and Q are inclined at an angle $\frac{\pi}{k}$. The value of *k* is

3. If $n^2+96: n\in\mathbb{N},$ is square of an integer then number of possible values of n is

4. If $2\cos\theta+\sin\theta=1,$ then maximum value of $7\cos\theta+6\sin\theta$ is

5. In $\Delta ABC,\ \angle A=\frac{\pi}{6},\ \angle B=\frac{\pi}{4}$. If $c^2=ka^2+b^2$ then value of k^2 is

Space for Rough work